CHAPTER 49

Sexual Dysfunction and Infertility

Male and Female Infertility

Infertility affects around 1 in 7 couples of reproductive age, often causing substantial psychological distress.

Causes of infertility in Women - Female factor-(35-40%)

A. May result from anovulation or from abnormalities of the reproductive tract that prevent fertilisation or embryonic implantation, most commonly damaged fallopian tubes from previous infection.

B. Ovulatory dysfunction
- Polycystic ovarian syndrome
- Hypergonadotrophic hypogonadism
- Hypogonadotrophic hypogonadism

C. Tubular dysfunction
- Pelvic inflammatory disease (chlamydia, gonorrhoea)
- Previous sterilisation
- Previous pelvic or abdominal surgery
- Endometriosis

D. Cervical and/or uterine dysfunction
- Congenital abnormalities
- Treatment for cervical
- Fibroids carcinoma
- Asherman's syndrome

E. Unexplained or mixed factor (20-35%)

Causes of infertility in Men - Male factor (35-40%)

1. It may result from impaired sperm quality (e.g. reduced motility) or number.
2. Azoospermia or oligospermia is usually idiopathic, but may be a consequence of hypogonadism
3. Microdeletions of the Y chromosome are increasingly recognised as a cause of severely abnormal spermatogenesis.
4. In many couples, more than one factor causing subfertility is present, and in a substantial proportion no cause can be identified.
5. Reduced sperm quality or production
   - Y chromosome microdeletions
   - Hypergonadotrophic hypogonadism
   - Varicocele
   - Hypogonadotrophic hypogonadism
6. Tubular dysfunction
   - Varicocoele
   - Previous sexually transmitted infection (chlamydia, gonorrhoea)
   - Congenital abnormality of vas deferens/ epididymis
   - Previous vasectomy
7. Unexplained or mixed factor (20-35%)

Clinical assessment

1. A history of previous pregnancies, relevant infections and surgery is important in both men and women.
2. A sexual history must be explored sensitively as some couples have intercourse infrequently or only when they consider the woman to be ovulating, and psychosexual difficulties are common.
3. Irregular and/or infrequent menstrual periods are an indicator of anovulatory cycles in the woman, in which case causes such as PCOS should be considered.
4. In men, the testes should be examined to confirm that both are in the scrotum and to identify any structural abnormality, such as small size, absent vas deferens or the presence of a varicocoele.

Investigations

1. Investigations are generally performed after a couple has failed to conceive despite

unprotected intercourse for 12 months, unless there is an obvious abnormality (e.g. amenorrhoea).

2. The male partner should provide at least two fresh semen samples, over an interval of several weeks, for analysis of sperm count and quality.
3. In women with regular periods, ovulation can be confirmed by an elevated serum progesterone concentration on day 21 of the menstrual cycle.
4. In women with irregular periods, such as in PCOS, weekly urine samples can be collected over several months to detect any peaks in progesterone metabolites which indicate ovulation.
5. Transvaginal ultrasound can be used to assess uterine and ovarian anatomy.
6. Tubal patency may be examined at laparoscopy or by hysterosalpingography (HSG; a radio-opaque medium is injected into the uterus and should normally outline the fallopian tubes).
7. In vitro assessments of sperm survival in cervical mucus may be ascertained in cases of unexplained infertility but are rarely helpful.

Male hypogonadism

1. The clinical features of both hypo- and hypergonadotrophic hypogonadism include loss of libido, lethargy with muscle weakness, and decreased frequency of shaving.
2. Patients may also present with gynaecomastia, infertility, delayed puberty and/or anaemia of chronic disease.

Investigations

1. Male hypogonadism is confirmed by demonstrating a low serum testosterone level.
2. The distinction between hypo and hypergonadotrophic hypogonadism is by measurement of random LH and FSH.

Gynaecomastia

1. Gynaecomastia is the presence of glandular breast tissue in males.
2. Gynaecomastia results from an imbalance between androgen and oestrogen activity, which may reflect androgen deficiency or oestrogen excess.
3. The most common are physiological: for example, in the newborn baby (due to maternal and placental oestrogens), in pubertal boys (in whom oestradiol concentrations reach adult levels before testosterone) and in elderly men (due to decreasing testosterone concentrations).
4. Prolactin excess alone does not cause gynaecomastia
5. Hypogonadism, liver failure, oestrogen secreting tumour, Drugs (digoxin, cimetidine etc) etc may cause gynaecomastia.

Hirsutism

1. Hirsutism refers to the excessive growth of thick terminal hair in an androgen-dependent distribution in women (upper lip, chin, chest, back, lower abdomen, thigh, forearm) and is one of the most common presentations of endocrine disease.
2. It should be distinguished from hypertrichosis, which is generalised excessive growth of vellus hair..

Semen analysis, Semen Examination and Semen Counting Procedure

Sample

- This is preferred if the patient in the lab for the collection of the sample.
- Masturbation is preferred, and the entire collected semen should be submitted.
- The accepted volume is 2 to 5 mL.
- Collect the sample when the doctor or the technician should be available to evaluate the motility immediately.
- 2 to 3 days of sexual abstinence is preferred.
- Prolonged abstinence should be discouraged because of the sperm cells' quality, and especially their motility will decrease.
- The proper container is also important.
- To evaluate the vasectomy, the patient needs to ejaculate once or twice before the semen specimen collection day. This process will

clear the distal portion of the vas deference.

- Samples are unsatisfactory if the sample is obtained at the house by coitus interrupts or masturbation.
- In the house sample, instruct the patient to deliver it within one hour of collection.
- Instruct the patient to avoid cold and heat during transportation. The best is to keep the sample in the closed hands (in the fist).

Precautions

- Don't use condoms, particularly with spermicide.
- The specimen should be maintained at 37 ^{0}C during transport if brought from home and should be examined within 3 hours of collection.
- A sterile container is needed, and the sample should be collected at room temperature of 37 ^{0}C.
- Plastic containers are not recommended.
- Avoid extreme temperatures.
- The analysis should be done immediately when the semen is liquefied.
- Should be examined within 4 hours,
- The sample should be kept at 37 ^{0}C.
- Wait till liquefaction is complete for the examination.

Drugs that may decrease the count are

- Antineoplastic agents (nitrogen mustard, procarbazine, vincristine, and methotrexate).
- Direct the patient to avoid alcohol intake for several days before the semen collection.
- Instruct the patient to avoid sexual activity for 2 to 3 days before collecting the sample.
- Avoid prolonged abstinence because the quality of the sperms and their motility may be decreased.

Purpose of the Test (Indications)

- This is the workup of the infertile couple.
- This is done to evaluate the quality of sperms.
- To confirm the efficacy of vasectomy (postvasectomy).
- Done for the forensic studies.
- To get semen for artificial insemination

Pathophysiology

Definition of semen

- Seminal fluid is the secretion of prostate glands, two seminal vesicles, two Cowper glands, and testes serve as the vehicle for sperms transport and create the alkaline medium. It also provides nutrition. This serves as the medium which promotes the sperm's motility and survival.
- Semen (seminal fluid) consists of a combination of products of various male reproductive organs like testes, epididymis, seminal vesicle, prostate, bulbourethral, and urethral glands.
- The ejaculated semen is a viscous, yellow-grey fluid, which forms a firm gel-like clot immediately after the ejaculation.
- This clotted semen liquefies at room temperature (37 ^{0}C) within 5 to 60 minutes. If it requires more than one hour, that semen is considered abnormal.

Semen analysis includes

- Volume.
- Sperm count.
- Evaluation of the normal sperm.
- Evaluation of sperm motility.
- The pH of the semen.

Spermatogenesis occurs in the seminiferous tubules of the testes. These are matured in the epididymis

- Semen production is dependent upon the function of testes.
- These are stimulated by testosterone.
- The concentration of testosterone in the testes is 100 times more than the peripheral blood.
- This high concentration of testosterone is needed for spermatogenesis.
- LH stimulates Leydig's cells to produce testosterone.
- These are lined with 46 chromosome germ cells called spermatogonia.

- These cells undergo mitotic division and ultimately give rise to primary spermatocytes.
- Secondary spermatocytes give rise to spermatids, and these will differentiate into sperm or spermatozoa containing 23 chromosomes
- Semen production depends upon the function of the testis.
- Hypothalamus producing a gonadotropin-releasing hormone (GRH) in response to low testosterone levels.
- GRH stimulates the pituitary gland to produce the follicular stimulating hormone.
- Now the FSH stimulates the Sertoli cell, which is the location of sperm production.
- FSH stimulates the Sertoli cells to secrete activin, which stimulates sperma-togenesis.
- Besides, the Sertoli cells secrete inhibin, which inhibits FSH secretion from the pituitary gland.
- LH stimulates the Leydig cells, which produce testosterone. Testosterone then stimulates the seminiferous tubules to produce sperm.
- Spermatogenesis (in the seminiferous tubules) takes approximately 3 months.
- Maturation of the spermatogonia to mature spermatocytes takes approximately 75 days, through the epididymis.
- In the epididymis, the motility is acquired; it takes another 14 days.
- Spermatocytes develop from the Sertoli cells in the seminiferous tubules. They get matured in the epididymis and are ready for fertilization.
- Hemospermia when there are abundant RBCs in the ejaculate.

In primary gonadal failure

- FSH is raised.
- LH is raised.

In secondary gonadal failure

- FSH is decreased.
- LH is decreased.

Classification of spermatocytes dependent upon the sperm count

- Aspermia (azoospermia) when there is no sperm.
- Oligospermia when the count is < 20 million/mL.
- Asthenospermia is low sperm motility.
- Necrospermia - Normal counts of the sperm but are non-motile.

Semen Analysis Includes

1. Volume. Check when the sample is fresh.
2. Sperm count. This can be done when the semen is completely liquified.
3. Motility. This should be checked when the sample is fresh and repeat when it is liquefied.
4. Morphology. This can be better appreciated if the smear is stained.

Normal semen

1. The volume is 2 to 6 ml.
2. Color is gray to white or opalescent.
3. Liquefaction time normally 10 to 30 minutes at 37 ^{0}C. Some other says 15-20 min.
4. The volume normally is 1.5 to 5 mL.
5. pH varies between 7.7 to 8. Below, 7.0 usually has prostatic secretion.
6. Sperm count is 20 to 250 million/mL.
   - Normally a total count of more than 80 million/ml. But now, it is considered as > 20 million /mL as fertile semen.
7. Live spermatozoa of more than 50%.
   - >65% of the sperm does not take stain, and these are alive.
8. Sperm motility >50% at one hour. This is moderate to the strong forward motion.
   - Place the semen drop on the prewarmed slide at 37 ^{0}C.
   - Examine under the 40x and find the motile sperm.
   - Evaluate the sperm actively motile and sluggish one.
   - Normally >70% of sperms are motile after one hour when the sperms are kept at 37 ^{0}C.

9. The progressive motility score is 3 to 4.
10. Sperm morphology should be >30%. Normally normal forms are >70%, and immature forms are <4%.
    - Can stain the slide with the Papanicolaou method.
    - Count at least 200 sperms under oil emersion.
    - Abnormal forms are <30%.
    - Note immature sperms as well.
    - Also, note RBCs and polys.
11. Microscopy:
    - WBC is 4 to 6/HPF.
    - Epithelial cells are few.
    - RBCs are not found.
    - Aggregation may be seen, but very few.
    - Crystals are amorphous phosphate.
12. Stain the slide with eosine, which will be taken up by the dead sperms only.
13. Clumping or agglutination of the sperms should be reported, which indicates antibodies.
14. The fructose level is 150 to 600 mg/dL.
    - Low or nil fructose indicates an obstruction in the ejaculatory duct.
    - The spermatozoa utilize fructose.
15. Acid phosphatase = 100 to 300 mg/100 ml.
16. Citric acid = >3 mg/ml.
17. Zinc = >75 µ/ml.
18. Magnesium = >70 µ/ml.
19. Inositol = >1 mg/ml.
20. Glucosidase = >20 mU/ ejaculate.
21. Carnitine = >250 µ/ml.

Source 2 Normal semen

1. Volume = 2 to 5 mL
2. Liquefaction time = 20 to 30 minutes afte the collection.
3. pH = 7.12 to 8.0
4. Sperm count = 50 to 200 million/mL
5. Sperm motility = 60 to 80% actively motile.
6. Sperm morphology = 70 to 90% normal shaped

Normal semen (another source)

Characteristic features	Values per ejaculate
The volume of the semen per ejaculate	>2 mL
Sperm density	>20 million/mL
Sperm count	>40 million/mL
Motility	1. >50% forward progression 2. >25% rapid progression within 60 minutes of ejaculation
Morphology	>30% normal morphology
pH	7.2 to 8.0
Color (appearance)	Gray-white to yellow
Time of liquefaction	Within 30 minutes
Fructose level	>1200 µ/mL
Acid phosphatase	100 to 300 mg/mL
Citric acid	>3 mg/mL
Zinc	>75 µ/mL
Magnesium	>70 µ/mL
Inositol	>1 mg/mL
Carnitine	>250 µ/mL
Prostaglandins (PGE1 and PGE2)	30 to 200 µ/mL
Glucosidase	>20 mU per ejaculate
Glycerylphosphory-lcholine	>650 µ/mL

Pathological Semen

1. The color is brown to red. or may be yellow and turbid.
2. Liquefaction time is >60 minutes.
3. Suppose the volume is <1.5 ml. This maybe <2 ml or >5 ml.
4. pH is <7.2.
5. Total count <20 million /mL.
6. Live sperm <35 %.
7. Motility <50 % after 2 hours.

8. The progressive motility score is 0 to 1.
9. Morphology >30 % abnormal forms.
10. Fructose is decreased.

Difference between the normal and abnormal semen:

Parameter	Normal semen	Patholog-ical semen
Coagulation	Coagulates	Delayed
Liquefaction	Complete in 10 to 30 minutes	Delayed
pH	7.2 to 7.8	
Volume	2 to 6 mL	<1.5 mL
Sperm density x 106/mL	>20	<10
Total sperm count x106/mL/ per ejaculate	>80	<20
Progressive motility score (after 2 to 4 hours of ejaculate)	3 to 4	0 to 1
Live spermatozoa	≥ 50%	<35%
One hour after ejaculate	≥ 70%	
After 3 hours of ejaculate	≥ 60%	
After 4 hours of ejaculate	>50%	<35%
Normal spermatozoa	≥ 60%	<35%
Defective heads	<35%	>60%
Defective mid-piece	≥ 20%	>25%
Defective tails	≥ 20%	>25%
Immature forms	<4%	

The first method to count Sperms

1. **Sperm count solution:**
   - Formalin = 1 mL
   - Sodium bicarbonate = 5 grams
   - Distilled water = 100 mL
   - Mix well the above ingredients.
   - Add all the above ingredients into 100 mL distilled water.
2. **Procedure for counting the sperms:**
   - Take 0.02 ml semen
   - Take 0.38 mL above solution (diluting fluid).
   - Mix well above semen and diluting solution.
   - Now load the Neubauer chamber.
   - Count the large two squares.
3. **Important** that moves the microscope up and down to count the sperm at a different level.
4. **Calculation:** Count in the large two square x100,000 = count in million /mL

Formula to calculate the sperm count

Sperm count = Number of sperms in 2 large square (2 sq. mm) x 100,000=Count in million/mL

Counting procedure

- Improved Neubauer chamber (Haemocytometer) is used for counting
- The diluted semen is carefully mixed and the chamber is charged using a Pasteur pipette.
- The chamber is then examined by using x 10 objective of microscope.
- Sperms are counted in the four large corners and one large central square.

The second method to count sperms in the semen:

1. Dilute the semen with water or above sodium bicarbonate solution.
2. Take 1:20 dilution of the semen.
3. Count the five RBC counting squares (4 at the corner and one at the center).
4. **Calculations:** suppose the sperms are 50 in all these 5 RBC square.

   1.50 x 1,000,000 = 50,000,000/mL (million/mL)

Semen : Microscopic Analysis

- Sperm count
- NV = 20-160 million / mL
- Make 1 to 20 dilution with sodium

bicarbonate and fromalin, count 5 small squares (within the center large square) of the Neubauer hemacytometer.

To calculate % motility

1. Count at least 200 motile or non-motile sperms in 5 different microscopic fields. % of the nonmotile sperm are calculated as follows:

Formula to calculate the sperm motility

$$\text{Motility} = \frac{\text{Total sperms - Nonmotile sperms \%}}{\text{Total sperms}} \times 100$$

Normally, 50% or more sperms are motile

Sperm viability

- The ejaculated sperm in the semen remain viable for several days in the female genital tract. Fertilization can occur as long as the sperms are alive, and these can survive for 5 days.
- The sperms can be preserved for decades when the semen is freezed at -20 ^{0}C.
- Sperm viability is needed when the progressively motile sperms are less in number.

Method to assess the viability

1. Take one drop of the semen.
2. Add an equal volume of vital stain (trypan blue).
3. Cover with the cover glass.
4. Keep for a few minutes, but not more than 5 minutes.
5. Count 100 cells, including viable sperm (not stained) and dead sperm (take stain).
6. This will give you the count of viable sperms.

Forensic Value of Semen Analysis

1. **Acid phosphatase** is in high concentration in the semen.
2. It is suggested that a swab from the suspected case of rape is taken, preserve in 2.5 mL of a protective broth, and store at 4 ^{0}C or room temperature.
   - A sample stored in this way will keep acid phosphatase activity for one month.
3. In the sample with thymolphthalein as the substrate, it was found that vaginal acid phosphatase in noncoital women is <10 U/L.
   - There is a noncoital level if the vaginal sample is taken after 4 days of intercourse.
4. The sampling swab can be kept in 1 mL of normal saline, but these samples are freezed and thaw at 2 to 4 ^{0}C for 24 hours before the assay.
5. **Creatinine phosphokinase** (CPK) is a high level in the semen. This can also be used for medicolegal purposes.

Importance of the Semen Analysis

1. That one sample cannot confirm the infertile couple because of variation in the count of semen.
2. To repeat one more sample. If possible, take the third sample at an interval of 3 weeks apart.
3. In case of vasectomy, no sperms should be seen in the ejaculate.

Work up for the infertile male

1. Get the report of the previous semen analysis.
2. Advise antispermatozoal antibody.
3. Advise luteinizing hormone (LH) and Follicle-stimulating hormone (FSH). These hormones will give the effect of the pituitary gland on gonadal function and spermatogenesis.
4. Sims-Huhner test. In this test, a postcoital cervical mucus smear is taken. To see the sperm mobility penetration in the mucus and maintenance of the mobility.

Drugs that may decrease the Count are

1. Cancer chemotherapy (Vincristine, methotrexate, procarbazine, and nitrogen mustard).
2. Estrogen therapy.
3. Cimetidine.
4. Methyltestosterone.

Important Note

- Single semen analysis is not conclusive because the sperm count varies from day today.
- A semen analysis should be done twice or thrice for the best result

Sperm Antibody, Anti Sperm-antibody, Anti Spermatozoal antibody

Sample

1. This is done on the fresh semen of the patient and sample preferred by masturbation.
2. First, urinate and then clean and wash the penis before collecting the sample.
3. Keep the sample at body temperature if collected at home.
4. The sample cannot be collected by sexual intercourse because there will be contamination by vaginal secretion.
5. Take the blood of the male and female for serum preparation.
6. Collect cervical mucus from the female.

Purpose of the Test (Indications)

1. This is a screening test for an infertile couple.
2. This detects antibodies against the spermatozoa.

Pathophysiology

1. Sperm antibodies are directed against the sperm antigen.
2. Antisperm antibodies are found in:
   - Men with the blocked efferent duct.
   - Men with a vasectomy show a 30 to 70% antibody.
3. Mechanism of antibody formation.
   - Sperm from the blocked ducts gives rise to autoantibody formation as the immune system recognizes these as forgein.These antibodies are agglutinin or cytotoxic.
4. The antisperm antibody of IgA type is against the tail of sperm, and they inhibit the motility and decrease the penetration of cervical mucus.
5. Sperm antibody of IgG type blocks the sperm-ovum fusion.
6. A high titer of sperm antibodies is strong evidence of infertility.
7. Sperm antibodies are found in 20% of the female. Their presence in the female significance is not clear because some of these ladies may become pregnant.
8. In some studies :

   1.Women with primary infertility are 75% positive for sperm agglutinin antibodies.

   2.11 to 15% of pregnant ladies show the same level of sperm antibodies.

Normal

- Negative.

Anti-Sperm Antibodies are seen in:

1. Blockage of testicular efferent ducts.
2. After the vasectomy, almost every male has an antisperm antibody. After the reversal of the vasectomy antisperm, the antibody disappears.
3. Infertility. This may be seen in both males and females.

Treatment

For to treat antisperm antibodies, the possibilities are:

1. Use of the condom.
2. Sperms are processed to remove the antibodies.
3. Immunosuppressive therapy can be given both in males and females.
4. Invitro fertilization (IVF) can be tried.

Differential diagnosis of infertility and sexual dysfunction

S.No.	Infertility	Sexual dysfunction
1	Infertility is the failure to conceive (regardless of cause) after 1 year of unprotected intercourse.	Disturbed sexual intercourse & pleasure
2	This condition affects approximately 10-15% of reproductive-aged couples.	Sexual dysfunction can be any problems that prevent a person or couple from experiencing satisfaction from sexual activity. Some 43% of women and 31% of men report some degree of sexual dysfunction.
3	**Female factors** that affect fertility include the following categories: **Cervical:** Stenosis or abnormalities of the mucus-sperm interaction **Uterine:** Congenital or acquired defects; may affect endometrium or myometrium; may be associated with primary infertility or with pregnancy wastage and premature delivery **Ovarian:** Alteration in the frequency and duration of the menstrual cycle-Failure to ovulate is the most common infertility problem **Tubal:** Abnormalities or damage to the fallopian tube; may be congenital or acquired **Peritoneal:** Anatomic defects or physiologic dysfunctions (eg, infection, adhesions, adnexal masses)	**Female factors** **Desire disorders:** lack of sexual desire or interest in sex. **Arousal disorders:** inability to become physically aroused or excited during sexual activity. **Orgasm disorders:** delay or absence of orgasm (climax). **Pain disorders:** pain during intercourse. **Lack of vaginal lubrication:** Inadequate vaginal lubrication before and during intercourse. **Vaginal muscles:** Inability to relax the vaginal muscles enough to allow intercourse.
4	**Male factors:** **Pre-testicular:** Congenital or acquired diseases of the hypothalamus, pituitary, or peripheral organs that alter the hypothalamic-pituitary axis **Testicular:** Genetic or nongenetic **Post-testicular:** Congenital or acquired factors that disrupt normal transport of sperm through the ductal system	**Male factors:** • Inability to achieve or maintain an erection (hard penis) suitable for intercourse (erectile dysfunction). • Absent or delayed ejaculation despite enough sexual stimulation (retarded ejaculation). • Inability to control the timing of ejaculation (early, or premature ejaculation).

MCQs

1. Infertility is when a couple fails to conceive after
   a. 4 years of unprotected sex
   b. 3 years of unprotected sex
   c. 2 years of unprotected sex
   d. 1 year of unprotected sex
2. In women, this sexually transmitted disease can result in infertility
   a. Human papillomavirus (HPV)
   b. Pelvic inflammatory disease
   c. Genital herpes
   d. All of the above

Answer

1. d 2. b